Dana Naderi

PhD Researcher in Statistics
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EDUCATION

PhD in Statistics, CEREMADE, Université Paris Dauphine–PSL, Paris, France 2024–Present
Supported by a French Ministry for Europe and Foreign Affairs (MEAE) scholarship.

Master's Degree in Data Science, Allameh Tabataba'i University, Tehran, Iran 2020–2023
GPA 17.88/20 (4.0/4.0).

Bachelor's Degree in Mathematics, Farhangian University, Tehran, Iran 2013–2017
GPA 17.56/20 (3.78/4.0).

RESEARCH INTERESTS

Bayesian inference & computation | Monte Carlo methods | High-dimensional data analysis | Probabilistic machine learning

RESEARCH EXPERIENCE

PhD Researcher 2024–Present
CEREMADE, Université Paris Dauphine–PSL Paris, France

- Bayesian computation, Monte Carlo methods and marginal likelihood (evidence) estimation; joint work with C. P. Robert, K. Kamary and D. Wraith.

Master's Research 2020–2023
Allameh Tabataba'i University Tehran, Iran

- Bayesian networks, Bayesian penalized regression for high-dimensional data, and nonparametric regression for unstructured data.

SOFTWARE

ECMLE, *R package, author and maintainer* 2026

- Implements the Elliptical Covering Marginal Likelihood Estimator for approximating model evidence; published on CRAN (v0.1.0, March 2026) with documentation, reproducible examples and tests.

PUBLICATIONS

- **Naderi, D., Robert, C. P., Kamary, K. & Wraith, D. (2026).** Approximating evidence via bounded harmonic means. *Statistics and Computing*, 36(3), Article 120.
- **Decoding Neural Signals: Invasive BMI Review** (2025). In *Modern Technologies in Healthcare* (1st ed., pp. 151–211), CRC Press. With R. Firuzi, A. Bokani, J. Hassan, H. Ahmadyani, M. F. Abdi & D. Ebrahimi.
- **Bayesian Penalized Regression in High-Dimensional Data Analysis: A Case Study on Raman Spectroscopy for Disease Diagnosis** (2023). TechRxiv preprint. With F. Eskandari & M. Mahdavi.
- **Nonparametric Regression Estimation under Kernel Polynomial Model for Unstructured Data** (2022). *Journal of Statistical Research of Iran.* With F. Eskandari, S. Naghizadeh Ardabili & M. Mahdavi.

PRESENTATIONS

- **Approximating evidence via bounded harmonic means.** Contributed talk, MCQMC 2026 (17th International Conference on Monte Carlo and Quasi-Monte Carlo Methods in Scientific Computing), University of Edinburgh, UK, 9 June 2026.
- Same work presented at the **Mostly Monte Carlo Seminar**, PariSanté Campus, Paris (5 June 2026) and the **CEREMADE Young Researchers Seminar**, Paris (21 May 2026).
- **Nonparametric Regression Estimation under Kernel Polynomial Model for Unstructured Data.** 16th Iranian Statistics Conference, University of Mazandaran, Babolsar, Iran, 24–26 August 2022.

TEACHING EXPERIENCE

High-School Mathematics Teacher 2017–2022
Education Department Marivan, Kurdistan, Iran

TECHNICAL SKILLS

Programming: R, Python | **Software:** R package development (CRAN), Git/GitHub
Methods: Bayesian inference and computation, Monte Carlo methods, regression and nonparametric regression, high-dimensional data analysis, probabilistic graphical models
Tools: Linux, LaTeX, Microsoft Office

LANGUAGES

English (upper-intermediate) | French (intermediate) | Kurdish (native) | Farsi (fluent)